HYDRAULIC ENGINEERING 424 – 2026 Group Assignment

Meerlustkloof River – Hydrodynamic System Analysis Assignment

• Given: 20 February 2026 (Friday)
• Due: 16 March 2026 (Monday) @10:00
• Oral presentations: 17 March 2026 (Tuesday) @ 14:00 – 17:00
• Submission: STEMLearn (PDF) + Turn-It-In (open 13 March, closes 16 March at 9:00)

***Assignment support session – Wednesday, 25 February @10:00 to 12:00, M3002.**

This assignment will be conducted in groups. Group members are responsible for assigning roles within the team. A peer review will be conducted at the end of the project.

Overview

You are required to prepare a one-dimensional steady-flow hydrodynamic model of a reach of the Meerlustkloof River near Villiersdorp using HEC-RAS.

This assignment requires you to:

- Determine the 1:100-year flood lines.
- Analyse hydraulic behaviour of the system.
- Evaluate how the system responds to realistic development and environmental scenarios.
- Demonstrate engineering judgement and modelling reasoning.

This is an analysis and evaluation task, not a detailed design exercise. Bridges are to be ignored in this assignment. Appendix A shows a satellite image of the river and photographs of the river.

Given Files

• A contour map of the relevant river reach, both in pdf and AutoCAD formats (“Meerlustkloof Contours.pdf” & “Meerlustkloof Contours.dwg”);
• KMZ files indicating locations:
- Chainage 3905m: “Station 0 (Chainage 3905m).kmz”
- Chainage 0m:
    - Right bank flood plain: “Chainage 0m (Station 3905m) Right Bank Floodplain.kmz”
    - Top of the right bank: “Chainage 0m (Station 3905m) Right Bank Top.kmz”

• Excel spreadsheet (“Meerlustkloof.Info¡xlsx”) containing:
- Cross-sections at 500 m intervals
- River centre-line coordinates

- Projection file and shapefile, and GeoTiff terrain file (“Meerlustkloof_Projection.prj, “Meerlustkloof_Projection.shp”& Meerlustkloof_Geo.tif”)

PART A – BASELINE MODEL

1) Cross-Section Completion

Note at Chainage 0m (Station 3905m) no contour information is provided between the right bank floodplain and the top of the right bank. Your task is to extend the cross section provided in “Meerlustkloof Info.xlsx” by doing the following:

- Importing the *.kmz files into Cape Farm Mapper (available at https://gis.elsenburg.com/apps/cfm/)
- The distance between the right bank flood plain and top of the right bank is approximately 90 m. Therefore, your task is to measure the ground elevation between the two points.
- See hints on how to import the *.kmz files into Cape Farm Mapper and how to draw an elevation profile

2) HEC-RAS Model Set-Up

Set up a Hec-Ras model of the section of the Meerlustkloof River from chainage 0 m to chainage 3905 m (station 3905m to station 0m respectively).

- Use RAS-Mapper to import the projection file and GeoTiff file and use the GIS map to define the river reach of the Meerlustkloof River.
  - Assume a main channel roughness with Manning n = 0.04; and on the floodplains n = 0.06. Figure 2 and 3 depicts the main channel and floodplain roughness.
  - Use the Meerlustkloof 100-year flood peak as your upstream boundary condition. Assume at chainage 0 m (station 3905 m) normal flow conditions arise. Use a bed slope of $S_0 = 0.0215$.
  - The Elandskloof River joins the Meerlustkloof River at chainage 1500 (station 2405 m) (Figure 1). Include an additional point source (flow change location) to accommodate the discharge of the Elandskloof River.
  - Assume normal flow conditions as your downstream boundary condition. Use a bed slope of $S_0 = 7.25 \times 10^{-3}$.

- When running the model, select the proper flow regime (supercritical, subcritical or mixed). Provide the motivation for the selection in your report.

Annual Recurrence Interval (years)	**Meerlustkloof flood peak (m^3/s)**	**Lower Elandskloof combined flood peak (m^3/s)**
100	375	550

Include the following in your report:

- Downstream Reach Lengths (in a table format) used for each station.
- All the input values for your HAS-RAS Steady Flow simulation.
- GIS Map obtained from RAS-Mapper which includes the terrain, the location of the river reach as well as the location of the cross sections.

3) Baseline Floodline Determination

Use the results of your HEC-RAS simulation to predict the 1:100-year flood lines of the Meerlustkloof River. Plot the floodlines on the contour drawing in the given AutoCAD file, note that the floodlines are based on energy levels, not the water levels.

Include the following in your report:

- The cross sections obtained from HEC-RAS at chainage 0m, 1500m and 3905m (Stations 3905m, 2405m and 0m, respectively) which includes the 1:100-year water levels, energy levels etc. of the river at these chainages.
- The profile plot generated from HEC-RAS which includes the 1:100-year water and energy levels.
- Images of your AutoCAD drawing showing the floodlines, cross-sections and contours.
- RAS Mapper depth and velocity plots of 1:100-year flood peak (not required to import in AutoCAD).

4) Baseline Hydraulic Behaviour Analysis

Analyse and quantify:

- Maximum water surface elevation and its location
- Maximum velocity and its location
- The hydraulic effect of the Elandskloof confluence
- Land use types/infrastructure at risk

Your discussion must include quantified values and hydraulic reasoning.

PART B – SCENARIO-BASED SYSTEM ANALYSIS

Each group will be assigned one fictional planning scenario.

You must:

1. Interpret the physical implications of the scenario.
2. Decide how the hydraulic model should be modified (i.e. which parameters to adjust, by how much, and why).
3. Clearly justify your modelling assumptions, use relevant published work to inform your choices e.g. design guidelines, research reports, journal articles, etc.
4. Re-run the model and compare results with the baseline case.

All modifications must be implemented in the steady 1D HEC-RAS framework. There is no single correct parameter adjustment. Marks are awarded for reasoning, justification and hydraulic interpretation.

1) Scenario Options

Scenario 1 – Informal Settlement Development (Groups 1, 2,3)

Agricultural workers establish an informal settlement on the right bank floodplain between chainage 1000 m and 2000 m. Vegetation is removed, and areas are levelled to create building platforms, reducing effective floodplain storage and altering surface roughness.

Scenario 2 – Climate Intensification (Groups 4, 5, 6)

Regional climate projections for the Western Cape indicate change in the intensity of extreme rainfall events. Recent studies suggest that design storm magnitudes for rare events may increase over the coming decades.

Scenario 3 – Riverside Tourism (Groups 7, 8, 9)

Several boutique lodges and associated infrastructure are proposed along the banks of the Elandskloof River near its confluence with the Meerlustkloof River. To accommodate these developments, floodplain areas are levelled and raised to create building platforms and access routes.

Scenario 4 – Floodplain Rehabilitation (SUDS-Type Intervention) (Groups 10, 11, 12)

A floodplain rehabilitation project is proposed along sections of the study reach. Existing cultivated areas are converted into vegetated wetland corridors and indigenous riparian vegetation is re-established along the channel and floodplain margins.

2) **Required Outputs for Scenario Analysis**

For your assigned scenario, include:

- Overlayed longitudinal profiles (baseline vs scenario)
- Comparison table showing:
    - Maximum water surface elevation
    - Maximum velocity
    - Change in flood extent
    - Clear explanation of hydraulic mechanisms driving change

PART C – CRITICAL REFLECTION

In approximately 300-400 words, reflect on:

- Which factor most strongly influences system behaviour
- Whether the system appears resilient or sensitive
- Limitations of steady-flow modelling
- The modelling assumption introducing greatest uncertainty

PART D – ORAL PRESENTATION

Each group will deliver a **5-minute presentation, followed by 5 minutes Q&A** during tutorial session.

Your presentation must include:

- Baseline system summary
- Explanation of scenario modification
- Quantified comparison of results
- Statement on system resilience

Maximum 8 slides.

All group members must be prepared to answer any questions related to all aspects of the project.

REPORT REQUIREMENTS

- Cover page, including your group number, names, student numbers, date, title
- Table of group members roles and responsibilities
- Table of contents; List of Figures, List of Tables
- Summary of your model set-up (assumptions, boundary conditions etc.) for baseline and scenario.
- Results and interpretation, discussion
- Conclusions
- Appendix with floodlines drawing: contours, sections and floodlines, and other relevant information.

Maximum length: 25 pages (excluding TOC, appendix, etc).

ASSESSMENT CRITERIA

Marks will be awarded for:

- Correct model development
- Depth of hydraulic interpretation
- Quality of scenario justification
- Quantitative comparison
- Engineering reasoning
- Professional presentation

Descriptive reporting without analysis will not receive high marks. See marking rubric.

The HEC-RAS software is available on all FIRGA PCs. Download HEC-RAS for personal devices:

https://www.hec.usace.army.mil/software/hec-ras/download.aspx

Unfortunately, HEC-RAS is officially designed for Windows, and there is no native macOS version. But you can still run HEC-RAS on a Mac using the following method:

- Install Parallels Desktop or VMware Fusion to run Windows on your Mac.
- Install Windows within the VM.
- Download and run HEC-RAS as you would on a regular Windows PC.

APPENDIX A

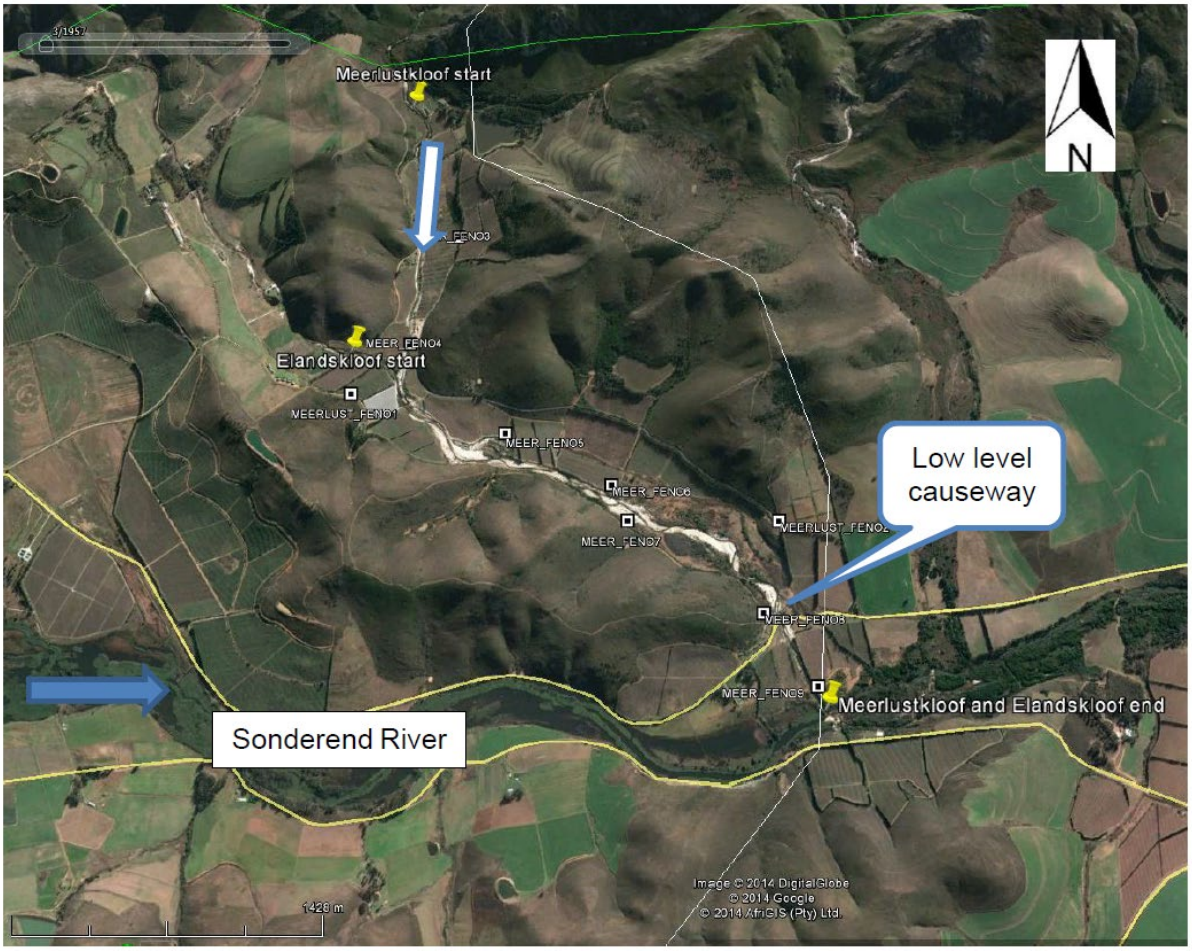

Figure 1: Study reach with the Sonderend River in the south.

Figure 2: Meerlustkloof River

Figure 3: Low-level causeway as viewed from upstream.